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Multimodal Interpretable Depression Analysis Using Visual, Physiological, Audio and Textual Data

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Abstract

Motivated by depression's significant impact on global health, this work proposes MultiDepNet, a novel multimodal interpretable depression detection system integrating visual, physiological, audio, and textual data. Through dedicated feature extraction methods (MTCNN for video, TS-CAN for physiological, ResNet-18 for audio, and RoBERTa for text modalities) and a strategic fusion of modality-specific networks including CNN-RNN, Transformer, MLP, and ResNet-18, it achieves significant advancements in depression detection. Its performance, evaluated across four benchmark datasets (AVEC 2013, AVEC 2014, DAIC, and E-DAIC), demonstrates average MAE of 5.64, RMSE of 7.15, accuracy of 74.19%, precision of 0.7373, recall of 0.7378, and F1 of 0.7376. It also implements a MultiViz-based interpretability mechanism that computes each modality's contribution to the model's performance. The results reveal the visual modality to be the most significant, contributing 37.88% towards depression detection.

1. Introduction

Depression is a mental health disorder characterized by persistent sadness and disinterest in daily activities [22, 39]. It is a major cause of disability globally and is linked to issues like chronic pain, anxiety, heart disease, and memory problems [11, 38]. Its early detection is essential for effective treatment [19, 35]. Traditional methods for detecting depression, like interviews and self-reports, can be subjective and biased, leading to potential misdiagnoses [45]. Machine learning based methods offer an objective alternative by analyzing quantitative data such as facial videos, nonverbal cues, and head movements [23, 24]. They reduce bias through systematic analysis of video, audio, and text data [29], with remote photoplethysmography (rPPG) enhancing non-invasive monitoring [49, 56].

We propose MultiDepNet, a multimodal depression detection system that uses Multi-task Cascaded Convolutional Network (MTCNN), Multi-Task Temporal Shift Attention Network (TS-CAN) [26], ResNet-18, and Robustly Optimized BERT Approach (RoBERTa) to extract visual, rPPG, audio, and textual features, respectively. These features are processed by CNN-RNN, Transformer, ResNet-18 and MLP, respectively. Multimodal features are combined using early and late fusion techniques, with an interpretability method based on MultiViz [25] to evaluate each modality's contribution to the determined depression class.

MultiDepNet has been evaluated across four datasets: AVEC 2013 [53], AVEC 2014 [54], DAIC [13], and E-DAIC [43], showing a mean absolute error (MAE) of 5.64, root mean squared error (RMSE) of 7.15, accuracy of 74.19%, precision of 0.7373, recall of 0.7378, and an F1-score of 0.7376. These results match or surpass the current state-of-the-art, highlighting its effectiveness in multimodal depression detection. The interpretability module determined visual modality to be the most influential at 37.88%, while rPPG, audio, and textual data contributed 22.70%, 21.17%, and 18.25% respectively. The absence of visual and rPPG modalities significantly reduced system performance, in contrast to the lesser impact of missing audio and textual data. This work's key contributions are as follows:

- We propose a multimodal depression recognition system, MultiDepNet, that combines visual, physiological, audio, and textual data using early and late fusion techniques.
- Specialized networks have been employed for each modality, ensuring precise feature extraction and classification for depression analysis.
- A MultiViz-based interpretability mechanism to identify individual contributions of each modality to the overall depression detection outcome has been incorporated.

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- MultiDepNet has been evaluated on AVEC 2013, AVEC 2014, DAIC, and E-DAIC datasets using MAE, RMSE, Accuracy, Precision, Recall, and F1 score, offering a more comprehensive analysis than previous studies.

2. Related Works

2.1. Unimodal Depression Analysis

Visual analysis for depression detection has advanced from basic feature extraction to sophisticated deep learning approaches. In this direction, Valstar et al. initially used facial expressions and body language [53,54]. This was followed by Yu et al.'s exploration of cloud-edge collaborative models [60], and Zhou et al.'s introduction of interpretable models with DepressNet [63]. Unimodal depression analysis was furthered by De et al. [7] and Zhu et al. [64] by using CNNs, and it was further enhanced by He et al. and Pandya et al. [16, 37] by incorporating attention mechanism.

Research into physiological patterns, including rPPG signals, remains relatively unexplored in depression analysis. Notable developments include Fan et al.'s introduction of a transformer-based framework that incorporates rPPG signals to enhance depression detection accuracy [12], and Casado et al.'s demonstration of significant precision improvements using unsupervised machine learning regressors for physiological data [5].

In the context of audio analysis for depression detection, Niu et al. introduced a spatiotemporal attention network and an attention feature fusion strategy, enhancing the representation of depression cues [33]. This approach, coupled with CNNs and LSTM networks, has been further developed by Gupta et al. and Han et al. [14, 15].

Text-based depression analysis has evolved with notable works by Chen et al. and Zheng et al. highlighting the role of text in identifying depression-related language patterns [6,62]. Extensions of this analysis to social media content by Dou et al. and Tejaswini et al. have enabled early depression detection and intervention possibilities [10, 50].

2.2. Multimodal Depression Analysis

Integrating multimodal data has significantly advanced depression analysis, offering more accurate diagnoses [42, 53, 54]. Kaya and Salah explored visual and audio modalities [20], while Zhu et al. and Jan et al. contributed dual-channel visual analysis and feature embedding, respectively [18,64]. Zhou et al.'s visual ensembling [63] and Pan et al.'s multimodal approach [36] further enriched the field. In the DAIC dataset, He et al. used bidirectional GRU networks for audio analysis [17], complemented by Sun et al.'s audio spectrogram integration with transformer networks [48]. Rodrigues et al. and Uddin et al. used CNN-LSTM framework and spatiotemporal network on the E-DAIC dataset that utilized textual and audio data [44, 52].

While these advancements typically utilize two modalities, only two methods have been identified in the literature that use three modalities: Casado et al., combining visual, audio, and physiological data [5], and the E-DAIC baseline employing visual, audio, and textual modalities [43]. In contrast, the proposed system, MultiDepNet, integrates all four modalities—visual, audio, textual, and physiological—thereby enhancing depression detection across diverse datasets evaluated using five evaluation metrics. It also incorporates an interpretability mechanism to compute each modality's contribution to detecting depression levels.

2.3. Interpretability for Depression Analysis

Recent efforts have integrated interpretability techniques for depression and emotion detection, enhancing model interpretability for mental health analysis [4, 21]. Notable contributions include the development of interpretable classification models by Burdisso et al. [4] and the selection of key behavioural cues by Alghowinem et al., along with Zhou et al. with DepressNet [2, 63]. Despite these advancements, existing studies do not incorporate interpretability analysis simultaneously for visual, physiological, audio and textual modalities in depression analysis—our work is the first to do so. We employ a MultiViz-based interpretability approach to quantify contributions from these modalities.

3. MMDepNet

The proposed system, MultiDepNet, is diagrammatically depicted in Fig. 1, where the individual's eyes have been obscured to ensure anonymity. The feature extraction models we utilize—MTCNN for video, TS-CAN for physiological data, ResNet-18 for audio, and RoBERTa for text—are selected for their robust feature extraction abilities in various domains. MTCNN excels in facial detection, crucial for analyzing emotional expressions in visual data [61]. TS-CAN is adept at identifying physiological signals linked to emotional distress [58], while ResNet-18 captures audio variations that can indicate mood changes [31]. RoBERTa effectively analyzes textual content, uncovering underlying emotional states [28].

Problem Formulation: To formalize the problem of multimodal depression detection, we define a multimodal dataset $\mathcal{D} = \{(\mathbf{V}_i, \mathbf{A}_i, \mathbf{T}_i, \mathbf{P}_i, y_i)\}_{i=1}^{N}$, where $\mathbf{V}_i$, $\mathbf{A}_i$, $\mathbf{T}_i$, and $\mathbf{P}_i$ represent the visual, audio, textual, and physiological features of the i-th subject in a collection of N instances, and y_i is the corresponding label indicating the level of depression. Our goal is to learn a predictive function $f : \mathcal{D} \rightarrow \mathcal{Y}$, where $\mathcal{Y} = \{0, 1, 2, 3\}^N$ represents the set of labels for depression classes: none, mild, moderate, and severe. The function f effectively leverages the information contained within and across the different modalities to predict the accurate depression level y_i.

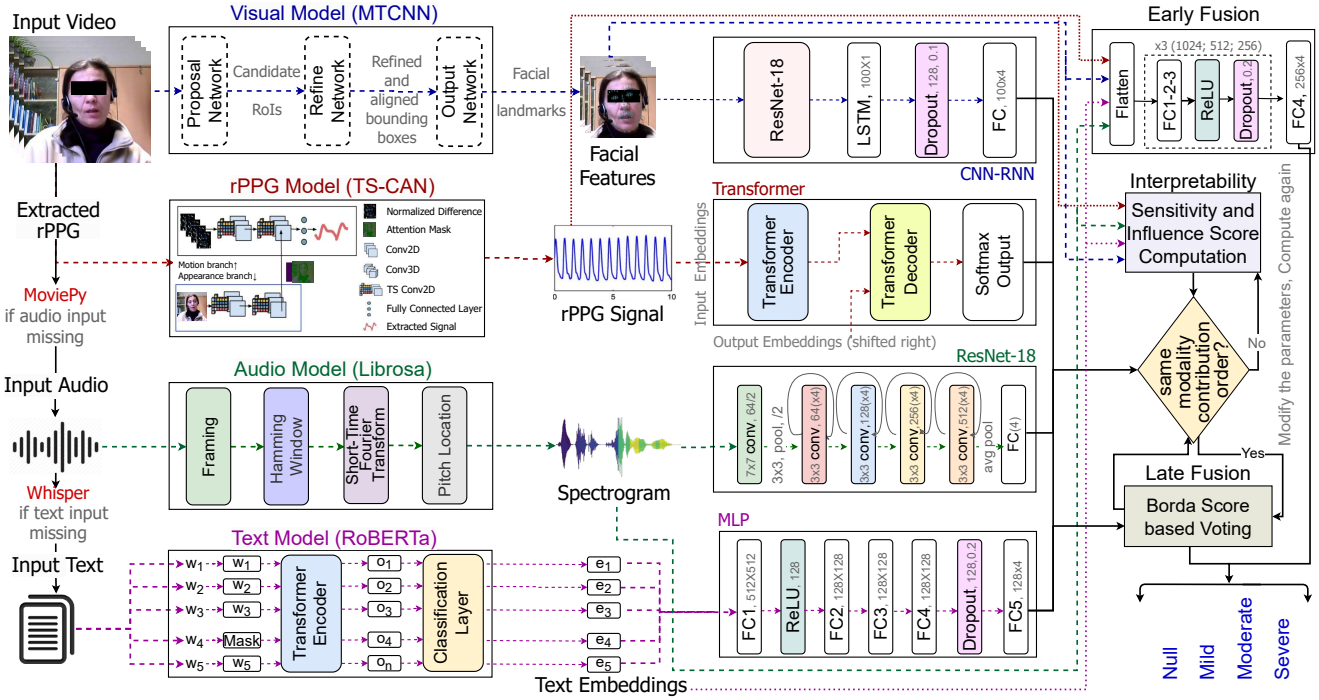


Figure 1. Schematic architecture of MMDepNet, illustrating the feature extraction and classification networks for visual, physiological, audio, and textual modalities, including interpretability and fusion modules for enhanced multimodal depression detection.

3.1. Visual Module

3.1.1 Visual Feature Extraction

We employ MTCNN [61] to perform robust face detection and alignment. This procedure is executed through a three-stage cascading function consisting of P-Net (Proposal Network), R-Net (Refine Network), and O-Net (Output Network), which progressively refine detection and alignment results in a cascading manner, as delineated in Eq. 1. This method processes video data by extracting and cropping facial regions, which are then resized to 224×224 pixels to ensure uniformity, thus facilitating further analysis.

$$P_i = P_{i-1} * N_i(I; \Theta_i) \tag{1}$$

where P_i represents the prediction at stage i, N_i denotes the network at stage i, I is the input image, and Θ_i are the parameters of the i-th network.

3.1.2 Visual Network

The visual network is designed to exploit the spatial and temporal information from video sequences effectively. This is achieved through a synergistic architecture combining convolutional neural network (CNN) for spatial feature extraction and recurrent neural network (RNN), specifically a long short-term memory (LSTM) network, for temporal dynamics analysis. This CNN-RNN architecture is preferred over other video classification models such as FaceNet [46] because it leverages the strengths of both CNNs for spatial and LSTMs for temporal analysis.

CNN Component: We use ResNet-18 for feature extraction due to its effective balance of complexity and performance. Pre-trained on ImageNet [9] containing 14 million images, ResNet-18 leverages learned features that are fine-tuned for depression detection. It extracts high-dimensional feature vectors from each video frame, essential for detecting subtle visual indicators of depression. Given an input frame $v_t \in \mathbb{R}^{H \times W \times C}$ at time t, the ResNet-18 model processes the frame as detailed in Eq. 2.

$$F_t = \text{ResNet-18}(v_t; \Theta_{\text{cnn}}), \tag{2}$$

where $F_t \in \mathbb{R}^D$ is the extracted feature vector with dimension D, and Θ_{cnn} denotes the parameters of the ResNet-18 model. The feature extraction module is designed to capture spatial dependencies within the frame, encoding critical information about facial expressions and other visual cues related to depression.

RNN Component: For modelling the temporal dependencies across the video sequence, the extracted features F_t are fed into an LSTM network, which is configured to process sequences of features, allowing it to capture temporal patterns such as changes in facial expressions over time that indicate depressive behaviour. Let $F_1, F_2, \ldots, F_T$ be the sequence of feature vectors for a video with T frames. The LSTM processes this sequence as per Eq. 3.

$$H_t, (C_t, M_t) = \text{LSTM}(F_t, (C_{t-1}, M_{t-1}); \Theta_{\text{lstm}}) \tag{3}$$

where $H_t \in \mathbb{R}^K$ represents the hidden state of the LSTM at time t, K being the output dimensionality, and (C_t, M_t) the cell and memory states; Θ_{lstm} denote the LSTM parameters. This LSTM configuration excels at analyzing long-term temporal patterns indicative of depression. The final classification stage processes the last timestep output H_T through a fully connected layer with dropout for regularization, then a softmax activation to classify depression severity – none, mild, moderate, severe.

3.2. Physiological Module

3.2.1 rPPG Feature Extraction

Utilizing the TS-CAN approach as detailed in [26], we employ an automated face detection algorithm to isolate facial regions from video frames. For a video $V = \{v_t | t = 1, ..., T\}$ with T frames, each frame v_t undergoes face detection, and the identified facial region f_t is resized to standard dimensions to ensure uniform input to our analysis model. We calculate the mean intensity within the facial region of interest (ROI) to generate the rPPG signal for each video, formalized as per Eq. 4.

$$\bar{I}_c = \frac{1}{N} \sum_{x=1}^{W} \sum_{y=1}^{H} I_{x,y,c} \tag{4}$$

where $I_{x,y,c}$ represents the intensity at pixel (x, y) for colour channel c, and N, W, and H are the total number of pixels, width, and height of the ROI, respectively. Following extraction, pixel values within the green channel are normalized, and the ROI is subdivided to capture spatial variations in blood volume, enhancing the signal's quality. This incorporation of spatial subdivisions and channel normalization reflects TS-CAN's focus to enhance data quality through spatial attention within network architecture.

3.2.2 rPPG Network

A Transformer network [55], has been employed to classify rPPG signals. The architecture includes: 1) An embedding layer to convert the input to a 64-dimensional space. 2) A positional encoding layer with a dropout rate of 0.1, enhancing sequence information. 3) A Transformer encoder with two layers, each with 64-dimensional size and 8 attention heads, tailored for sequential rPPG data processing. 4) A dropout layer at a rate of 0.5 for regularization, followed by a fully connected layer that projects the 64-dimensional representation into four depression classes (none, mild, moderate, severe). The network processes rPPG signals through the encoder, applies a mean operation across the sequence for a unified representation, and classifies depression levels. The Transformer's selection for rPPG signal classification stems from its proficiency in capturing long-range dependencies within temporal sequences, crucial for accurately identifying subtle, time-sensitive patterns in rPPG data, addressing long-distance frame forgetting [57].

3.3. Audio Module

3.3.1 Audio Feature Extraction

Audio features for depression detection are derived using the librosa library [30] to generate spectrograms, essential for capturing the prosodic features of speech, such as tone, pitch, and rhythm. The Short-Time Fourier Transform is applied to audio signals $x(t)$ to obtain a spectrogram S, defined as per Eq. 5.

$$S(f, t) = \int x(t) w(t - \tau) e^{-j2\pi f \tau} d\tau, \tag{5}$$

where $w(t)$ is a window function. Audio data from videos is extracted using the moviepy library [65], allowing consistent analysis across datasets. Additionally, the Whisper model by OpenAI facilitates the processing of non-English audio, integrating it into our multimodal analysis framework [34].

3.3.2 Audio Network

The audio network employs ResNet-18, adapted for classifying depression levels from audio signals. It processes the spectrogram S to output a feature vector F_a, capturing audio characteristics relevant to depression as per Eq. 6.

$$F_a = R_\psi(|S(f, \tau)|) = \text{ResBlocks}(\text{Conv2D}(|S(f, \tau)|)), \tag{6}$$

where Conv2D and ResBlocks denote the initial convolutional and residual block layers, respectively. This architecture utilizes skip connections, enhancing deep learning models by optimizing feature extraction for nuanced depression detection through audio. The features F_a are then processed by fully connected layers for final classification.

3.4. Text Module

3.4.1 Text Feature Extraction

In the process of extracting textual features from non-English audio, we utilize a two-phase approach involving OpenAI's Whisper model for speech recognition [34]. Whisper, leveraging a Transformer architecture, converts audio A into transcribed text $W(A)$ by processing log-Mel spectrograms to capture the audio's time-frequency characteristics. The transcribed text $W(A)$ is then input into RoBERTa, which processes the entire text as a single input, transforming it into a high-dimensional, fixed-size embedding vector E_t, as per Eq. 7.

$$E_t = \text{RoBERTa}(W(A)) \tag{7}$$

where E_t represents the fixed-size text embeddings obtained from $W(A)$ through the RoBERTa embedding mechanism. This fixed-size vector captures the semantic value of the transcribed text, ensuring consistent and effective processing across different inputs.

3.4.2 Text Network

For text classification, we employ a 5-layer Multi-Layer Perceptron (MLP) to classify the embeddings E_t into four depression levels. The MLP processes embeddings E_t through a series of transformations as per Eq. 8

$$F_t = L_5(L_4(L_3(L_2(L_1(e_t))))), \tag{8}$$

where F_t represents the final output of the MLP, providing a feature vector ready for classification. The output feature vector F_t is passed to a classification layer, which maps it to depression class probabilities using a softmax function as per Eq. 9.

$$C_t = \text{softmax}(W_c F_t + b_c), \tag{9}$$

where W_c and b_c denote the weight matrix and bias for the classification layer, and C_t, represents the probability distribution over depression classes. The network is trained to minimize the cross-entropy loss between the predicted probabilities C_t and the true class labels, optimizing the parameters of the MLP and the classification layer.

3.5. Multimodal Fusion

3.5.1 Early Fusion

For the early fusion, we preprocess and scale each modality's features for uniform data representation. Visual features $\mathbf{V}$ are normalized using z-score normalization. Audio features $\mathbf{A}$, represented as spectrograms, are standardized so each frequency bin has zero mean and unit variance. Textual features $\mathbf{T}$ are tokenized and then embedded via a pretrained model to ensure uniform embedding sizes. Physiological features $\mathbf{P}$ are preprocessed to include the calculation of the mean intensity $\bar{I}_c$ from the rPPG signals within the ROI, which captures relevant physiological information for the analysis. The preprocessed and scaled features from all modalities are concatenated before entering the model, enabling the network to learn intricate representations indicative of depressive states. This concatenated multimodal data is processed through a neural network that starts with a flattening layer for the multidimensional input tensor. It continues with a sequence of CNN layers, adapted for the increased input size from the fusion of the four data streams, and concludes with a final classification layer. The procedural flow is illustrated in Eq. 10.

$$\begin{aligned} I'_{\text{multi}} &= \text{preprocess_and_scale}(V, A, T, P, \bar{I}_c) \\ I''_{\text{multi}} &= \text{concatenate}(I'_{\text{multi}}) \\ I'''_{\text{multi}} &= \text{flatten}(I''_{\text{multi}}) \\ F_{\text{early-multi}} &= \text{NNet}(I'''_{\text{multi}}, C) \end{aligned} \tag{10}$$

where $\bar{I}_c$ is the mean intensity from the rPPG signals within the ROI, V denotes the visual features, A the audio features, T the textual embeddings, and P the physiological features. C symbolizes the number of classes for depression detection, and $F_{\text{early-multi}}$ the resultant feature vector from this multimodal fusion. $NNet$ is an experimentally determined neural network used for early fusion containing a combination of flattened, fully connected, Rectified Linear Unit (ReLU) and dropout layers as shown in Fig. 1.

3.5.2 Late Fusion

In the late fusion phase, the significance of each modality is evaluated using MultiViz [25], an interpretability method that utilizes feature attribution and interactive interfaces to explain how different modalities influence multimodal model outputs. This approach aids in identifying and analyzing each modality's impact. The Local Interpretable Model-agnostic Explanations (LIME) [41] technique has been used in the background of MultiViz, which provides explanations for predictions by approximating the local decision boundary of the model, helping to derive feature importance scores for each modality. These scores are aggregated by summing or averaging, then normalized to sum to one, serving as modality weights in the fusion process. Predictions from visual, physiological, audio and textual modalities are combined using these weights and the Borda count method, a consensus-based voting system that ranks options based on their relative standing across different evaluations, for dataset samples. With N modalities and M samples, we assign Borda scores to predictions for each sample j, calculated as shown in Eq. 11.

$$B_{ij} = \sum_{k=1}^{N} w_k \cdot R_{ik}, \qquad S_i = \sum_{j=1}^{M} B_{ij} \tag{11}$$

where R_{ik} denotes the rank of the i-th prediction from the k-th modality for the j-th sample, w_k represents the normalized weight for the k-th modality, and S_i represents the total score for the i-th prediction. This methodology ensures comprehensive consideration of predictions across all modalities, weighted by their estimated importance from the feature importance scores. To effectively handle instances of missing modalities, we incorporate an availability indicator a_{kj} for each modality k of each sample j, where $a_{kj} = 1$ signifies the presence and $a_{kj} = 0$ the absence of the modality. This inclusion allows for the adjustment of the Borda score computation to reflect only the available modalities. Consequently, the total score for each prediction is recalculated with these adjusted scores as per Eq. 12.

$$B'_{ij} = \sum_{k=1}^{N} a_{kj} \cdot w_k \cdot R_{ik}, \qquad S'_i = \sum_{j=1}^{M} B'_{ij} \tag{12}$$

The final decision for each sample is based on the highest recalculated total score S'_i, reflecting the dynamic contribution of available modalities weighted by their significance.

3.6. Interpretability

The MultiViz [25] method allows for probing both local and global effects of features on predictions, complemented

by a sensitivity analysis that highlights the direct influence of features on the output. Mathematically, the contributions of a feature f are quantified as per Eq. 13.

$$\text{Contrib}(f) = \alpha \cdot \text{Sensitivity}(f) + (1 - \alpha) \cdot \text{Influence}(f) \quad (13)$$

where Sensitivity(f) quantifies the local perturbation effects of feature f and Influence(f) represents the gradient-based influence of f on the model's predictions. The coefficient α serves as a balance factor between these two aspects. This formulation allows for a nuanced understanding of each feature's impact, guided by both local and global insights derived from the model's behaviour.

The interpretability module evaluates how individual features affect model performance, which is crucial for understanding their impact on predictive accuracy. If the importance of modalities, as indicated by Borda count scores in our late fusion strategy, does not align with their computed contributions, we adjust the interpretability model parameters and repeat the analysis. This iterative process involves fine-tuning the balance between sensitivity and influence to ensure that the contributions reflect true modality importance as demonstrated by the fusion outcomes.

4. Experimental Results

4.1. Datasets

MultiDepNet has been evaluated on four datasets summarized in Table 1 and described as follows: 1) Audio Video Emotion Challenge (AVEC) 2013 [53] featuring 105 videos, segmented into 30-second clips, yielding 1007 segments after undersampling to preserve class distribution for training. 2) AVEC 2014 [54] including 300 videos, equally divided between freeform and northwind recordings, all used in training. 3) Distress Analysis Interview Corpus (DAIC), or 'DAIC-Wizard-of-Oz' [13], consisting of 189 sessions segmented into 30-second clips, with 1229 segments utilized for training after undersampling. 4) E-DAIC [43] containing 275 sessions with 2787 segments. The DAIC and E-DAIC datasets lack raw video files, so only audio and textual modalities have been used. The four depression classes, based on Beck Depression Inventory II (BDI-II) [3], are none (0-13), mild (14-19), moderate (20-28), and severe (29-63).

Table 1. Summary of depression datasets

Name	Year	Size	Subjects
AVEC 2013 [53]	2013	150 Videos	58 Adults
AVEC 2014 [54]	2014	300 Videos	58 Adults
DAIC-WOZ [42]	2017	189 Sessions	189 Adults
Extended DAIC [43]	2019	275 Sessions	275 Adults

4.2. Training Strategy and Evaluation Metrics

The baselines and MMDepNet have been trained on an Nvidia V100 GPU, using a batch size of 16, a learning rate of 0.001, and 30 epochs, with 5-fold cross-validation and an 80%-20% train-validation split. Final evaluations used separate predefined test partitions—with 50, 100, 47, and 56 samples for AVEC 2013, AVEC 2014, DAIC, and E-DAIC, respectively—to verify performance on unseen data. The training utilized Adam optimizer, cross-entropy loss, and ReLU activation. Comprehensive metrics including MAE, RMSE, Accuracy, Precision, Recall, and F1-score provided a holistic model assessment, enhancing the evaluation depth compared to prior research.

4.3. Models

We evaluated MultiDepNet against several baselines, analyzing variations in visual and rPPG feature extraction and fusion techniques, crucial for depression detection. Baseline 1 utilizes Dlib for facial features and Haar cascades for rPPG signals with an early fusion approach. Baseline 2 mirrors Baseline 1 but employs a late fusion strategy with Borda scores. Baseline 3 replaces Dlib with MTCNN for visuals, maintaining Haar cascades and early fusion. Baseline 4 retains these methods and shifts to late fusion. Baseline 5 integrates TS-CAN for rPPG and MTCNN for visuals with early fusion. MultiDepNet, improving upon Baseline 5, adopts a Borda count-based late fusion, demonstrating enhanced performance as detailed in Table 2.

4.4. Results

Table 2 presents the accuracy of individual models on the AVEC 2014 dataset, where MMDepNet showed superior performance across other datasets as well. Due to space limitations, Tables 5, 7, and 6 focus on the AVEC 2014 dataset, while Tables 3 and 4 cover all datasets. These results highlight the effectiveness of the multimodal approach, with MMDepNet outperforming across datasets. This integration of multiple data modalities enhances the model's predictive power, demonstrating the robustness of the multimodal analysis in depression detection.

Table 2. Results for the baselines and proposed system. 'Acc,' 'P,' 'R' and 'F1' denote accuracy, precision, recall and F1 score. The ↑ or ↓ indicate that the metric's higher or lower values are better.

Model	MAE ↓	RMSE ↓	Acc ↑	P ↑	R ↑	F1 ↑
Baseline 1	8.64	11.91	51.38%	0.4852	0.5048	0.4948
Baseline 2	8.17	9.39	61.63%	0.6152	0.6079	0.6115
Baseline 3	8.13	10.11	58.05%	0.5726	0.5712	0.5719
Baseline 4	7.27	8.76	66.36%	0.6526	0.6561	0.6543
Baseline 5	7.97	8.99	62.77%	0.6162	0.6273	0.6217
MMDepNet	6.14	8.11	73.24%	0.7283	0.7313	0.7298

Interpretability Results: Table 3 shows each modality's average contributions to depression detection in early fusion. For AVEC 2013 and AVEC 2014, the visual modality contributes most significantly. For DAIC and E-DAIC, audio modality is the most prominent contributor.

Table 3. Analysis of modality-wise contribution.

Modality	AVEC2013	AVEC2014	DAIC	E-DAIC
Visual	37.43%	38.33%	–	–
rPPG	23.57%	21.83%	–	–
Audio	20.67%	21.67%	62.33%	58.83%
Textual	18.33%	18.17%	37.67%	41.17%

Comparison With State-of-the-art: Table 4 compares MMDepNet's results with state-of-the-art methods. Some methods excel on specific datasets, but struggle on others, while MMDepNet sustains good performance across all datasets. It ranks among top two for most metrics on AVEC 2013, AVEC 2014 and E-DAIC, and third in RMSE on AVEC 2014. Its lower performance on DAIC is due to lack of video feeds, limiting the use of visual and rPPG data.

4.5. Ablation Studies

4.5.1 Performance Analysis with Limited Modalities

Experiments with one, two, or three modalities have been conducted using MultiDepNet to assess their impact on depression detection. Early fusion substitutes zero vectors for missing modalities, allowing seamless data integration and learning. Late fusion adjusts to available modalities per sample, dynamically integrating only present data. Results from the AVEC 2014 dataset are detailed in Table 5, with consistent findings across other datasets. Performance improves significantly with the inclusion of visual and physiological data, while configurations relying solely on textual and audio data are less effective. Despite using four modalities, our system's training time per epoch is 2.07 times that of unimodal models, 1.3 times that of bimodal setups, and just 1.1 times longer than trimodal configurations. Even so, MMDepNet outperforms all unimodal, bimodal, and trimodal configurations in accuracy and other metrics, demonstrating a favourable trade-off between computational complexity and performance.

4.5.2 Choice of rPPG Feature Extraction Method

The choice of TS-CAN for our rPPG feature extraction is based on its performance in ablation studies. We tested various traditional and machine learning-based rPPG methods using the rPPG Toolbox [27]. As shown in Table 6, TS-CAN not only achieved the highest accuracy but also maintained the second-best signal-to-noise (SNR) ratio among all methods. This method significantly enhances rPPG signal quality by leveraging attention mechanisms that focus on temporal shifts, crucial for detecting subtle physiological changes associated with depression.

4.5.3 Choice of Late Fusion Technique

The three late fusion approaches explored are: 1) *Average Softmax Scores*: Averages softmax scores from multiple models for decision-making. 2) *Unsupervised Borda Count*: Aggregates class rankings from various modalities for consensus-based decision. 3) *Supervised Learning using MLP*: Adjusts fusion dynamically using backpropagation of classification errors. Experimental results in Table 7 show the efficacy of the Borda score-based voting for the AVEC 2014 dataset, implemented as our late fusion method. Similar observations have been noted across other datasets.

Table 4. Results' comparison with state-of-the-art. 'V,' 'P,' 'A,' and 'T' denote visual, physiological, audio, and textual modalities. 'LGBP-TOP,' 'SVR,' 'HOG,' and 'GCNN' refer to local dynamic binary patterns - three orthogonal planes, support vector regression, histogram of oriented gradients, and graph CNN, respectively. The best and second-best results are highlighted in bold and underlined. The ↑ indicates that higher values are better.

Method	Modality	MAE ↓	RMSE ↓
AVEC 2013			
Baseline paper [53]	VA	10.88	13.61
Local pattern analysis [59]	V	8.22	10.27
Eye-based feature extraction [20]	VA	7.86	9.72
Dual-channel analysis [64]	V	7.58	9.82
Depth-wise video analysis [1]	V	7.37	9.28
Two-stream spatiotemporal Net [51]	V	7.04	8.93
Unsupervised rPPG analysis [5]	V	6.43	8.01
Deep residual learning [7]	V	6.30	8.25
Multi-channel ensembling [63]	V	6.20	8.28
Depth-wise convolution analysis [32]	V	6.19	8.02
Two-stream 2DCNN [8]	V	**5.96**	<u>7.97</u>
MMDepNet	VPAT	<u>6.15</u>	**7.89**
AVEC 2014			
Base paper [54]	VA	8.86	10.86
Dual-channel analysis [64]	VA	7.47	9.55
Depth-wise video analysis [1]	V	7.22	9.20
Two-stream spatiotemporal Net [51]	V	6.86	8.78
Feature embedding network [18]	VA	6.68	8.04
Unsupervised rPPG analysis [5]	VPA	6.57	8.49
Multi-channel ensembling [63]	VA	6.21	8.39
Two-stream image analysis [8]	V	6.20	**7.94**
Deep residual learning [7]	V	**6.15**	8.23
Depth-wise conv analysis [32]	V	<u>6.14</u>	<u>7.98</u>
MMDepNet	VPAT	<u>6.14</u>	8.11
DAIC			
DAIC Baseline Multimodal [42]	VA	5.66	7.05
DAIC Baseline Unimodal [42]	V	6.12	6.97
PCA + HOG [47]	V	4.89	6.23
AVA-DepressNet Unimodal [36]	V	4.78	5.96
Linear Discriminant Analysis [40]	V	<u>4.64</u>	5.98
AVA-DepressNet Multimodal [36]	VA	**4.62**	<u>5.78</u>
Bidirectional GRU [17]	VA	<u>4.64</u>	**5.68**
MMDepNet	AT	4.99	6.27
E-DAIC			
Spectrogram + Transformer [48]	VA	-	8.67
Base paper [43]	AT	-	8.09
CNN [44]	A	-	7.02
LSTM [44]	T	-	6.88
CNN + LSTM [44]	AT	-	6.42
E-DAIC Baseline [43]	VAT	-	6.37
Spatio-Temporal Network [52]	AT	-	**5.77**
MMDepNet	AT	5.07	<u>6.31</u>

Table 5. Ablation study on missing modalities. Here, 'Acc,' 'P,' 'R' and 'F1' denote accuracy, precision, recall and F1 score whereas 'T' is per epoch training time. The ↑ or ↓ indicate that the metric's higher or lower values are better.

Model	T	MAE ↓	RMSE ↓	Acc ↑	P ↑	R ↑	F1 ↑
V	135s	8.25	11.06	63.64%	0.6412	0.6315	0.6363
P	117s	9.03	12.13	59.73%	0.5874	0.5783	0.5828
A	109s	8.97	12.11	60.17%	0.6106	0.5925	0.6014
T	87s	9.43	12.57	56.26%	0.5662	0.5524	0.5592
VP	178s	8.13	10.87	64.73%	0.6462	0.6517	0.6489
VA	171s	7.93	9.76	66.27%	0.6714	0.6767	0.6740
VT	157s	8.07	10.52	64.37%	0.6384	0.6319	0.6351
PA	162s	8.11	10.61	64.91%	0.6445	0.6464	0.6454
PT	144s	9.37	12.23	56.41%	0.5562	0.5598	0.5580
AT	139s	8.41	11.29	62.17%	0.6238	0.6214	0.6226
PAT	189s	7.78	10.13	65.17%	0.6513	0.6485	0.6499
VAT	197	7.94	9.26	66.42%	0.6583	0.6515	0.6549
VPT	201	7.23	9.27	68.57%	0.6912	0.6898	0.6905
VPA	210s	6.59	8.37	69.83%	0.6886	0.6912	0.6899
VPTA	232s	6.14	8.11	73.24%	0.7283	0.7313	0.7298

Table 6. Comparison of rPPG Feature Extraction Methods. The best and second-best values are highlighted in bold and underlined. The ↑ indicates that the metric's higher values are better.

Machine Learning methods			Traditional methods		
Method	**Acc ↑**	**SNR ↑**	**Method**	**Acc ↑**	**SNR ↑**
TS-CAN	**59.73%**	<u>25.2764</u>	POS	55.64%	**27.8321**
DeepPhys	57.87%	24.4981	ICA	52.18%	15.2743
PhysFormer	57.53%	20.7865	CHROM	51.43%	16.6412
PhysNet	<u>58.25</u>%	19.3157	BCG	49.33%	10.4583
Contrast-Phys	52.86%	13.4291	GREEN Channel	48.54%	11.1942
Face2PPG	50.27%	17.3174	Haar cascades	43.42%	10.1017

Table 7. Ablation study on late fusion approaches. Here, 'Acc,' 'P,' 'R' and 'F1' denote accuracy, precision, recall and F1 score. The ↑ or ↓ indicate that metric's higher or lower values are better.

Approach	MAE ↓	RMSE ↓	Acc ↑	P ↑	R ↑	F1 ↑
Avg. of softmax	7.51	10.21	58.09%	0.5745	0.5818	0.5781
Borda Score	6.14	8.11	73.24%	0.7283	0.7313	0.7298
Train using MLP	7.16	9.22	64.34%	0.6361	0.6423	0.6392

4.6. Discussion

MultiDepNet's salient features: MultiDepNet uses visual, physiological, audio, and textual data to enhance depression analysis. Using cameras to remotely capture physiological rPPG signals improves its convenience and applicability. It enhances the accuracy of depression detection by integrating these modalities, demonstrating how they complement each other in identifying emotional states. Late fusion, where modalities are integrated at a later stage, consistently outperforms early fusion, suggesting it is more effective in learning emotional states.

Observations from multimodal data: Our analysis across visual, physiological, audio, and textual data revealed the following observations: Subdued facial expressions and reduced blink rates have been noted in the visual domain, alongside crucial heart rate indicators linked to depression. These findings underline the significance of facial cues and heart rate in depression analysis. In scenarios without visual and physiological data, like phone calls, audio cues such as slower pitch rate and flattened tone become prominent. MMDepNet adapts to the available modalities, ensuring robust depression detection even with partial data. This flexibility highlights the importance of each modality in various contexts, enhancing our system's practical utility.

Potential Negative Societal Impact: While this work brings significant advancements in depression detection, challenges such as potential data sensitivity and overreliance on automated assessments need careful consideration. These issues are acknowledged in Section 5, where future efforts are described to focus on extending MultiDepNet for real-world applications while emphasizing privacy preservation and generalizability. This includes refining fusion techniques to improve data efficiency and integrating the technology with professional healthcare services, thus ensuring the system enhances rather than replaces human expertise in mental health assessments.

5. Conclusion and Future Plan

MultiDepNet advances depression analysis using a multimodal framework combining visual, physiological, audio, and textual data, emphasizing interpretability. The innovation lies in the effective fusion of these modalities, enabling detailed quantification of their contributions to depression detection. Emphasizing interpretability provides insights into each modality's contribution. We acknowledge potential privacy concerns due to the sensitive nature of the input data. Future efforts will focus on extending MultiDepNet for real-world applications, emphasizing privacy preservation and generalizability. The potential applications include telemedicine, wearable technology, and remote health monitoring systems, where multimodal data integration can significantly enhance diagnostic accuracy and patient care. We plan to refine fusion techniques to improve data efficiency and reduce model complexity. Collaborations among computational scientists, psychologists, and clinicians will further enhance depression analysis, extending to psychological health monitoring, student well-being, and workplace mental health interventions.

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References

[1] Mohamad Al Jazaery and Guodong Guo. Video-Based Depression Level Analysis By Encoding Deep Spatio-Temporal Features. *IEEE Transactions on Affective Computing*, 12(1):262–268, 2018.

[2] Sharifa Mohammed Alghowinem, Tom Gedeon, Roland Goecke, Jeffrey Cohn, and Gordon Parker. Interpretation of Depression Detection Models via Feature Selection Methods. *IEEE Transactions On Affective Computing*, 2020.

[3] Steer R. A. Beck, A. T. and O.K. Brown. Manual For The Beck Depression Inventory-ii. *The Psychological Corporation - San Antonio, TX*, 1996.

[4] Sergio Burdisso, Marcelo Errecalde, and Manuel Montes-y Gómez. A Text Classification Framework for Simple and Effective Early Depression Detection over Social Media Streams. *Expert Systems with Appl*, 133:182–197, 2019.

[5] Constantino Álvarez Casado, Manuel Lage Cañellas, and Miguel Bordallo López. Depression Recognition Using Remote Photoplethysmography from Facial Videos. *IEEE Transactions on Affective Computing*, 2023.

[6] Jian Chen, Yuzhu Hu, Qifeng Lai, Wei Wang, Junxin Chen, Han Liu, Gautam Srivastava, Ali Kashif Bashir, and Xiping Hu. IIFDD: Intra And Inter-Modal Fusion For Depression Detection With Multi-Modal Information From Internet Of Medical things. *Information Fusion*, 102:102017, 2024.

[7] Wheidima Carneiro De Melo, Eric Granger, et al. Depression Detection Based on Deep Distribution Learning. In *IEEE International Conference on Image Processing (ICIP)*, pages 4544–4548, 2019.

[8] Wheidima Carneiro De Melo, Eric Granger, et al. Encoding Temporal Information for Automatic Depression Recognition from Facial Analysis. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1080–1084, 2020.

[9] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A Large-Scale Hierarchical Image Database. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 248–255, 2009.

[10] Rongyu Dou and Xin Kang. TAM-SenticNet: A Neuro-Symbolic AI Approach For Early Depression Detection Via Social Media Analysis. *Computers and Electrical Engineering*, 114:109071, 2024.

[11] Timon Elmer and Christoph Stadtfeld. Depressive Symptoms Are Associated With Social Isolation In Face-To-Face Interaction Networks. *Scientific Reports*, 10(1):1444, 2020.

[12] Huiting Fan et al. Transformer Based Multimodal Feature Enhancement Networks For Multimodal Depression Detection Integrating Video, Audio and Remote PhotoPlethysmoGraph Signals. *Information Fusion*, 104:102161, 2024.

[13] Jonathan Gratch, Ron Artstein, Gale M Lucas, Giota Stratou, Stefan Scherer, Angela Nazarian, Rachel Wood, Jill Boberg, David DeVault, Stacy Marsella, et al. The Distress Analysis Interview Corpus Of Human And Computer Interviews. In *tHE 9th International Conference on Language Resources and Evaluation*, pages 3123–3128. Reykjavik, 2014.

[14] Sachi Gupta, Gaurav Agarwal, Shivani Agarwal, and Dilkeshwar Pandey. Depression Detection Using Cascaded Attention Based Deep Learning Framework Using Speech Data. *Multimedia Tools and Applications*, pages 1–39, 2024.

[15] Zhuojin Han, Yuanyuan Shang, Zhuhong Shao, Jingyi Liu, Guodong Guo, Tie Liu, Hui Ding, and Qiang Hu. Spatial-Temporal Feature Network For Speech-Based Depression Recognition. *IEEE Transactions on Cognitive and Developmental Systems*, 2023.

[16] Lang He, Jonathan Cheung-Wai Chan, and Zhongmin Wang. Automatic Depression Recognition Using CNN with Attention Mechanism from Videos. *Neurocomputing*, 422:165–175, 2021.

[17] Lang He, Mingyue Niu, Prayag Tiwari, Pekka Marttinen, Rui Su, Jiewei Jiang, Chenguang Guo, Hongyu Wang, Songtao Ding, Zhongmin Wang, et al. Deep Learning for Depression Recognition With Audio-Visual Cues: A Review. *Information Fusion*, 80:56–86, 2022.

[18] Asim Jan, Hongying Meng, Yona Falinie Binti A Gaus, and Fan Zhang. Artificial Intelligent System for Automatic Depression Level Analysis Through Visual and Vocal Expressions. *IEEE Transactions on Cognitive and Developmental Systems*, 10(3):668–680, 2017.

[19] DF Jhang, YS Chu, JH Cai, YY Tai, and CC Chuang. Pain monitoring using heart rate variability and photoplethysmograph-derived parameters by binary logistic regression. *Journal of Medical and Biological Engineering*, 41:669–677, 2021.

[20] Heysem Kaya and Albert Ali Salah. Eyes Whisper Depression: A CCA Based Multimodal Approach. In *22nd ACM International Conference on Multimedia (ACM MM)*, pages 961–964, 2014.

[21] Puneet Kumar, Sarthak Malik, Balasubramanian Raman, and Xiaobai Li. VISTANet: VIsual Spoken Textual Additive Net for Interpretable Multimodal Emotion Recognition. *arXiv preprint arXiv:2208.11450*, pages arXiv–2208, 2022. Accessed 10 July 2024.

[22] Puneet Kumar, Alexander Vedernikov, and Xiaobai Li. Measuring Non-Typical Emotions for Mental Health: A Survey of Computational Approaches. *arXiv preprint arXiv:2403.08824*, 2024. Accessed 10 July 2024.

[23] Xingyun Li, Xinyu Yi, Lin Lu, Hao Wang, Yunshao Zheng, Mengmeng Han, and Qingxiang Wang. TSFFM: Depression Detection Based On Latent Association Of Facial And Body Expressions. *Computers in Biology and Medicine*, 168:107805, 2024.

[24] Xingyun Li, Xinyu Yi, Jiayu Ye, Yunshao Zheng, and Qingxiang Wang. SFTNet: A Microexpression Based Method For Depression Detection. *Computer Methods and Programs in Biomedicine*, 243:107923, 2024.

[25] Paul Pu Liang, Yiwei Lyu, Gunjan Chhablani, Nihal Jain, Zihao Deng, Xingbo Wang, Louis-Philippe Morency, and Ruslan Salakhutdinov. MultiViz: Towards Visualizing and Understanding Multimodal Models. In *International Conference on Learning Representations (ICLR)*, 2023.

[26] Xin Liu, Josh Fromm, Shwetak Patel, and Daniel McDuff. Multi-Task Temporal Shift Attention Networks For On-Device Contactless Vitals Measurement. *Advances in Neural Information Processing Systems (NeurIPS)*, 33:19400–19411, 2020.

[27] Xin Liu, Girish Narayanswamy, Akshay Paruchuri, Xiaoyu Zhang, Jiankai Tang, Yuzhe Zhang, Roni Sengupta, Shwetak Patel, Yuntao Wang, and Daniel McDuff. rPPG-Toolbox: Deep Remote PPG Toolbox. *Advances in Neural Information Processing Systems (NeurIPS)*, 36, 2024.

[28] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A Robustly Optimized BERT Pretraining Approach. *arXiv preprint arXiv:1907.11692*, 2019. Accessed 10 July 2024.

[29] Karl Magtibay, Karthikeyan Umapathy, et al. A Review of Tools and Methods for Detection, Analysis, and Prediction of Allostatic Load due to Workplace Stress. *IEEE Transactions on Affective Computing*, 2023.

[30] Brian McFee, Colin Raffel, Dawen Liang, Daniel PW Ellis, Matt McVicar, Eric Battenberg, and Oriol Nieto. Librosa: Audio and Music Signal Analysis in Python. In *SciPy*, pages 18–24, 2015.

[31] Asif Iqbal Middya, Baibhav Nag, and Sarbani Roy. Deep Learning Based Multimodal Emotion Recognition Using Model-Level Fusion of Audio-Visual Modalities. *Knowledge-Based Systems*, 244:108580, 2022.

[32] Mingyue Niu et al. Multi-Scale & Multi-Region Facial Discriminative Representation For Automatic Depression Level Prediction. In *IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1325–1329, 2022.

[33] Mingyue Niu, Jianhua Tao, et al. Multimodal Spatiotemporal Representation for Automatic Depression Level Detection. *IEEE Transactions on Affective Computing*, 2020.

[34] OpenAI. Introducing Whisper. `https://openai.com/research/whisper`, 2023. Accessed 10 July 2024.

[35] Anastasia Pampouchidou, Matthew Pediaditis, et al. Automated Facial Video-Based Recognition of Depression and Anxiety Symptom Severity: Cross-corpus Validation. *Machine Vision and Applications*, 31(4):30, 2020.

[36] Yuchen Pan, Yuanyuan Shang, Zhuhong Shao, Tie Liu, Guodong Guo, and Hui Ding. Integrating Deep Facial Priors Into Landmarks For Privacy Preserving Multimodal Depression Recognition. *IEEE Transactions on Affective Computing*, 2023.

[37] Darshanaben Dipakkumar Pandya, Abhijeetsinh Jadeja, Sheshang Degadwala, and Dhairya Vyas. Diagnostic Criteria For Depression Based On Both Static And Dynamic Visual Features. In *International Conference on Intelligent Data Communication Technologies and Internet of Things*, pages 635–639. IEEE, 2023.

[38] Brenda WJH Penninx. Depression And Cardio-Vascular Disease: Epidemiological Evidence On Their Linking Mechanisms. *Neuroscience & Biobehavioral Reviews*, 74:277–286, 2017.

[39] Waleed Ragheb, Jérôme Azé, et al. Negatively Correlated Noisy Learners for At-Risk User Detection on Social Networks: A Study on Depression, Anorexia, Self-Harm, and Suicide. *IEEE Transactions on Knowledge and Data Engineering*, 35(1):770–783, 2021.

[40] Swati Rathi, Baljeet Kaur, and RK Agrawal. Enhanced Depression Detection From Facial Cues Using Univariate Feature Selection Techniques. In *International Conference on Pattern Recognition and Machine Intelligence*, pages 22–29. Springer, 2019.

[41] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "Why Should I Trust You?" Explaining the Predictions of Any Classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, pages 1135–1144, 2016.

[42] Fabien Ringeval, Björn Schuller, et al. AVEC 2017: Real-Life Depression, and Affect Recognition Workshop and Challenge. In *Proceedings of the 7th ACM International Workshop on Audio/visual Emotion Challenge*, pages 3–9, 2017.

[43] Fabien Ringeval, Björn Schuller, Michel Valstar, Nicholas Cummins, Roddy Cowie, Leili Tavabi, Maximilian Schmitt, Sina Alisamir, Shahin Amiriparian, Eva-Maria Messner, et al. AVEC 2019 workshop And Challenge: State-Of-Mind, Detecting Depression with AI, And Cross-Cultural Affect Recognition. In *Proceedings of the 9th ACM International on Audio/visual Emotion Challenge and Workshop*, pages 3–12, 2019.

[44] Mariana Rodrigues Makiuchi, Tifani Warnita, et al. Multimodal Fusion of Bert-CNN and Gated CNN Representations for Depression Detection. In *Proceedings of the 9th International on Audio/Visual Emotion Challenge and Workshop*, pages 55–63, 2019.

[45] Marco Sarchiapone, Carla Gramaglia, Miriam Iosue, Vladimir Carli, Laura Mandelli, Alessandro Serretti, Debora Marangon, and Patrizia Zeppegno. The Association Between Electrodermal Activity Depression And Suicidal Behaviour: A Systematic Review And Narrative Synthesis. *BMC Psychiatry*, 18(1):1–27, 2018.

[46] Florian Schroff, Dmitry Kalenichenko, and James Philbin. FaceNet: A Unified Embedding For Face Recognition And Clustering. In *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition (CVPR)*, pages 815–823, 2015.

[47] Bo Sun, Yinghui Zhang, Jun He, Lejun Yu, Qihua Xu, Dongliang Li, and Zhaoying Wang. A Random Forest Regression Method With Selected Text Feature For Depression Assessment. In *Proceedings of the 7th ACM Annual Workshop on Audio/Visual Emotion Challenge*, pages 61–68, 2017.

[48] Hao Sun, Jiaqing Liu, Shurong Chai, Zhaolin Qiu, Lanfen Lin, Xinyin Huang, and Yenwei Chen. Multi-modal Adaptive Fusion Transformer Network For The Estimation of Depression Level. *Sensors*, 21(14):4764, 2021.

[49] Toshiyo Tamura. Current Progress Of PhotoPlethysmoGraphy And SPO2 For Health Monitoring. *Biomedical Engineering Letters*, 9(1):21–36, 2019.

[50] Vankayala Tejaswini, Korra Sathya Babu, and Bibhudatta Sahoo. Depression Detection From Social Media Text Analysis Using Natural Language Processing Techniques And Hybrid Deep Learning Model. *ACM Transactions on Asian and Low-Resource Language Information Processing*, 23(1):1–20, 2024.

[51] Md Azher Uddin, Joolekha Bibi Joolee, and Young-Koo Lee. Depression Level Prediction Using Deep Spatio-Temporal

Features And MultiLayer Bi-LSTM. *IEEE Transactions on Affective Computing*, 13(2):864–870, 2020.

[52] Md Azher Uddin, Joolekha Bibi Joolee, and Kyung-Ah Sohn. Deep Multi-Modal Network Based Automated Depression Severity Estimation. *IEEE Transactions on Affective Computing*, 2022.

[53] Michel Valstar et al. AVEC 2013: The Continuous Audio/Visual Emotion and Depression Recognition Challenge. In *Proceedings of the 3rd ACM international workshop on Audio/visual emotion challenge*, pages 3–10, 2013.

[54] Michel Valstar, Björn Schuller, et al. AVEC 2014: 3D Dimensional Affect and Depression Recognition Challenge. In *Proceedings of the 4th International Workshop on Audio/Visual Emotion Challenge*, pages 3–10, 2014.

[55] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is All You Need. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 5998–6008, 2017.

[56] Alexander Vedernikov, Zhaodong Sun, Virpi-Liisa Kykyri, Mikko Pohjola, Miriam Nokia, and Xiaobai Li. Analyzing Participants' Engagement During Online Meetings Using Unsupervised Remote Photoplethysmography with Behavioral Features, 2024. Accessed 10 July 2024.

[57] Rui-Xuan Wang, Hong-Mei Sun, Rong-Rong Hao, Ang Pan, and Rui-Sheng Jia. TransPhys: Transformer-Based Unsupervised Contrastive Learning For Remote Heart Rate Measurement. *Biomedical Signal Processing and Control*, 86:105058, 2023.

[58] Wenjin Wang, Albertus C Den Brinker, Sander Stuijk, and Gerard De Haan. Algorithmic Principles Of Remote PPG. *IEEE Transactions on Biomedical Engineering*, 64(7):1479–1491, 2016.

[59] Lingyun Wen, Xin Li, Guodong Guo, and Yu Zhu. Automated Depression Diagnosis Based On Facial Dynamic Analysis And Sparse Coding. *IEEE Transactions on Information Forensics and Security*, 10(7):1432–1441, 2015.

[60] Yao Yu, Shuai Ding, et al. Cloud-Edge Collaborative Depression Detection Using Negative Emotion Recognition and Cross-Scale Facial Feature Analysis. *IEEE Transactions on Industrial Informatics*, 2022.

[61] Kaipeng Zhang, Zhanpeng Zhang, Zhifeng Li, and Yu Qiao. Joint Face Detection and Alignment Using MultiTask Cascaded Convolutional Networks. *Signal Processing Letters*, 23(10):1499–1503, 2016.

[62] Wenbo Zheng, Lan Yan, and Fei-Yue Wang. Two Birds With One Stone: Knowledge-Embedded Temporal Graph Network Using Clinical Notes for Patient-Level Clinical Diagnosis and In-Hospital Mortality Prediction. *Journal of Biomedical Informatics*, 125:104–116, 2023.

[63] Xiuzhuang Zhou, Kai Jin, Yuanyuan Shang, and Guodong Guo. Visually Interpretable Representation Learning For Depression Recognition From Facial Images. *IEEE Transactions on Affective Computing*, 11(3):542–552, 2018.

[64] Yu Zhu, Yuanyuan Shang, Zhuhong Shao, and Guodong Guo. Automated Depression Diagnosis Based On Deep Networks To Encode Facial Appearance And Dynamics. *IEEE Transactions on Affective Computing*, 9(4):578–584, 2017.

[65] Zulko. MoviePy: A Software Library For Video Editing. `https://github.com/Zulko/moviepy`, 2017. Accessed 10 July 2024.